

The Chain Rule

ANSWER KEY



Basic chain rule applications

1 Differentiate each function using the chain rule:

a $f(x) = (4x - 1)^5$
 $= 5(4x - 1)^4(4) = 20(4x - 1)^4$

b $f(x) = \sin(3x^2) = 6x\cos(3x^2)$

c $f(x) = e^{5x} = 5e^{5x}$

d $f(x) = \ln(2x + 7) = \frac{2}{2x + 7}$

2 Differentiate $f(x) = \sqrt{x^2 + 9}$.

$$f'(x) = \frac{2x}{2\sqrt{x^2 + 9}} = \frac{x}{\sqrt{x^2 + 9}}.$$

The chain rule with trig and exponential functions

3 Differentiate $f(x) = \cos^3(x)$ (i.e. $[\cos x]^3$).

$$f'(x) = 3\cos^2 x \cdot (-\sin x) = -3\cos^2 x \sin x.$$

4 Differentiate $f(x) = e^{\sin x}$.

$$f'(x) = \cos x \cdot e^{\sin x}.$$

5 Differentiate $f(x) = \tan(e^x)$.

$$f'(x) = \sec^2(e^x) \cdot e^x.$$

Nested (multiple) chain rule

6 Differentiate $f(x) = \sin(\sqrt{x^2 + 1})$, applying the chain rule twice.

$$f'(x) = \cos(\sqrt{x^2 + 1}) \cdot \frac{x}{\sqrt{x^2 + 1}}.$$

7 Differentiate $f(x) = (\ln(3x))^4$.

$$f'(x) = 4(\ln(3x))^3 \cdot \frac{3}{3x} = \frac{4(\ln(3x))^3}{x}.$$

Chain rule combined with product/quotient rules

8 Differentiate $f(x) = x^2 e^{3x}$.

$$f'(x) = 2xe^{3x} + x^2(3e^{3x}) = e^{3x}(2x + 3x^2).$$

9 Differentiate $f(x) = \frac{\sin(2x)}{x}$.

$$f'(x) = \frac{2\cos(2x) \cdot x - \sin(2x)}{x^2}.$$

Chain rule with tables of values

- 10 Given $f(2) = 5$, $f'(2) = 3$, $g(2) = 7$, $g'(2) = -1$, find $\frac{d}{dx}[f(g(x))]$ at $x = 2$ if $g(2) = 7$ and $f'(7) = 4$ (note the composition uses f' evaluated at $g(2) = 7$, not at 2).

$$\frac{d}{dx}f(g(x)) = f'(g(x)) \cdot g'(x); \text{ at } x = 2: f'(7) \cdot g'(2) = 4 \cdot (-1) = -4.$$

Chain rule with inverse trig and logarithms

- 11 Differentiate $f(x) = \sin^{-1}(2x)$.

$$f'(x) = \frac{2}{\sqrt{1-(2x)^2}} = \frac{2}{\sqrt{1-4x^2}}.$$

- 12 Differentiate $f(x) = \ln(\sin x)$.

$$f'(x) = \frac{\cos x}{\sin x} = \cot x.$$

- 13 Differentiate $f(x) = \tan^{-1}(x^2)$.

$$f'(x) = \frac{2x}{1+(x^2)^2} = \frac{2x}{1+x^4}.$$

Finding tangent lines using the chain rule

- 14 Find the equation of the tangent line to $f(x) = (2x - 1)^3$ at $x = 1$.

$$f(1) = 1^3 = 1. \quad f'(x) = 3(2x - 1)^2(2) = 6(2x - 1)^2; \quad f'(1) = 6(1)^2 = 6. \quad \text{Tangent: } y - 1 = 6(x - 1), \text{ i.e. } y = 6x - 5.$$